

Project Proposal:

A Data-Driven Study of Disney Films in Film Industry

Title and author

A Data-Driven Study of Disney Films in Film Industry, Runyu Fu

Research Questions

-How have the genres of Disney movies evolved over the years, and do these changes align with the overall trends in the global film and television industry?

-Which years were the most popular for Disney movie releases, and is this popularity connected to broader developments in the global movie market during the same periods?

-Are there any relationships between runtime, genre, IMDb rating, awards, and Metascore? (The analysis will not only explore whether these relationships are positive/negative, but also examine more detailed patterns and phenomena.)

Motivation

With the rapid development of society, people's lives have become increasingly rich and diverse, offering more ways to spend their time. Among these, film and television works have become a "necessity" in everyday life. Beyond individual hobbies, people are paying more and more attention to movies, TV series, and variety shows during their leisure time away from work and study. Therefore, in order to better understand and meet people's entertainment needs, it is necessary to study the development trends of film and television works and the key characteristics of popular productions in recent years.

Furthermore, as a major source of economic growth, analyzing the trends and audience preferences for different genres can also contribute to promoting the development of the entertainment economy. The reason I chose to focus specifically on Disney movies is that Disney holds a significant influence within the global entertainment industry. As a Disney fan myself, I also have a personal curiosity to explore this topic more deeply. I have noticed that Disney movies have remained well-received over time, and I want to understand which themes are most popular, when they started gaining global attention, and how these patterns relate to trends in the broader film and television industry.

Moreover, most current research in society tends to focus on scientific or academic topics, while relatively few studies take a comprehensive look at people's entertainment activities from both cultural and professional perspectives. Therefore, I believe this research is particularly meaningful. The findings may even provide insights that help guide adjustments in the film and television industry, ultimately contributing to a higher level of public happiness and cultural engagement.

Data Setting

Main dataset: Disney movies&shows <https://www.kaggle.com/datasets/eshummalik/disney>

The dataset collects information about Disney shows (show names, types, release dates, ratings, directors, genre, awards, etc.), which was collected by a user/collector called Eshum_malik on Kaggle. According to the dataset, the three contexts that may potentially influence my analysis are: the missing data in early years (the shows lack complete information in earlier years, so this may cause a different result); the judgement of genre changes overtime (which means is rather subjective and not uniform); confidential business compact (rates or scores may not be purely rated by audience).

Additional dataset: IMDB movies (top 1000)

<https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>

For this dataset, it collects information about movies and tv shows that are within the top 1000. The content contains are similar to the Disney dataset: show names, release dates, ratings, directors, genre, votes, etc.), which was collected by a user/collector called Harshit Shankhdhar on Kaggle. According to the dataset, the three contexts that may potentially influence my analysis are: the given condition (top 1000) data, which cannot fully represent movies all over the world; the time period covers a wide range, which have lots of influential factors to comparison, such as techniques; confidential business compact (rates or scores may not be purely rated by audience), similar as Disney dataset.

Challenging goals

1. Messy data

Both datasets above contain lots of missing values. Also, for some column titles, such as genre, there are multiple answers to it, which is hard to directly get them out by using simple codes, not to mention those with short descriptions. I have to think about how I can get the main words of those short sentences, and summarize as a dictionary or others.

2. Statistical Hypothesis

This project will involve multiple statistical hypothesis tests to explore the relationships among various variables, such as IMDb ratings, Metascores, and playback durations. I will conduct multiple statistical hypothesis tests—including *t-tests*, *correlation analyses*—to examine whether Disney movies differ significantly from the IMDb Top 1000 dataset in terms of *IMDb rating*, *Metascore*, and *runtime*. Each test will be formulated with explicit null and alternative hypotheses, and I will discuss the assumptions required for valid inference (e.g., normality and independence). The results will be interpreted in the context of film industry trends, allowing for evidence-based insights into how Disney productions compare with broader cinematic standards. Based on the results, I will revisit and refine my hypotheses, and, drawing on domain knowledge in film studies, provide deeper interpretations of the findings.

Method

Libraries that may be used: pandas, numpy, matplotlib, seaborn

Overview of steps:

1. Identify columns that the content inside needs to be revised in a uniform format
2. Create new columns, lists, or dictionaries in the dataset to facilitate data manipulation and plotting
3. Writing codes to get plots and models to study relationships between variables
4. Statistical tests & analysis

data-manipulation

- create a small dataset as `testing_csv` being used to test data manipulation
- revise 'released_year' and 'released_at' columns to numeric and comparable formats.
- remove empty spaces and 'N/A' values.
- convert 'released_at' to separate year/month/day columns using string splitting.
- clean 'IMDb rating', 'Meta score', and 'award' columns; split 'award' column into lists.
- get average, max, min (basic statistical elements) by using pandas library
- perform correlation analysis and other statistical tests using numpy and pandas as needed.

plotting function

- plot those relationships of imdb rating, meta score, and runtime out by using seaborn and matplotlib, for instance using `sns.relplot()`.
- create bar plots and catplots to show the distribution of movies per genre over time (10-year intervals)- label x as years and label y as counts of genre, `sns.catplot()`.
- use `sns.lmplot()` to show the imdb rating changes over time (check if the rating is growing over time)

Plan

task 1: data setting & cleaning (data manipulation)

task 2: data reorganizing with standard columns (data manipulation)

task 3: additional statistical values and calculations (data manipulation)

task 4: various plots (plotting function)

task 5: statistical test and interpretations

task 6: final analysis and report

Citations

Now, I still don't have any specific sources that are being used before getting the exact project out.